Lecture 22 — RREF and linear dynamical systems

Wed Oct 21

Last updated: October 23, 2026.

Boyd video (optional): Lecture 31 — solving linear equations · Lecture 26 — linear dynamical systems · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: RREF (class notes), Lachniet Ch. 2 (linear-system commands), and Boyd 9.1–9.3. Related objectives: Lectures. Schedule. Continues elimination from Lecture 21.

One meeting for RREF and the first dynamics lecture. Board the warm-up, the fill-ins, and **W1**–**W3**. **W4**–**W5** are the same ideas if the room is fast. Iteration and long-run patterns are Lecture 23 on Friday.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Read a reduced row echelon form: name pivot and free variables and write a parametric solution.
2. Tell inconsistent, unique, and infinitely many solutions apart from that form.
3. Write an evolving system as $x_{t+1} = Ax_t$ and build the transition matrix from a flow description.
4. Compute and interpret the next state or two.

Checking a software solve with the residual $Ax - b$ is mentioned below and practiced in Lab 2 and Lab 3. Many time steps, plots, and long-run patterns wait for the simulation lecture.

Warm-up

The RREF of an augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Which variables are pivots? Which are free? Write the general solution.

Fill in

Reduced row echelon form (RREF) is echelon form with these extras:

- Each pivot is _____.
- Each pivot is the _____ nonzero in its column (zeros above and below).

A **pivot variable** corresponds to a pivot column. A **free variable** has _____ pivot in its column; assign a parameter (for example $x_3 = t$).

Reading the RREF of $[A \mid b]$:

Pattern	Conclusion
A row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$	_____
Every column of A has a pivot, and the system is consistent	_____ solution
Consistent, and at least one free variable	_____ solutions

A row of zeros **including** the right-hand side is redundant, not inconsistent. Software can compute RREF or a solution (`rref`, `A\b`, `numpy.linalg.solve`). Verify a candidate x by the residual $Ax - b$, which should be near _____.

A **linear dynamical system** (discrete time) has the form

$$x_{t+1} = _____,$$

where $t = 0, 1, 2, \ldots$ is the time index.

- The **state** x_t is an n-vector that describes the system at time t.
- The matrix A is the **transition** matrix. Its size is _____.

Columns. Column j of A is the next state if the current state is _____.

A few steps by hand. $x_1 = Ax_0$, $x_2 = Ax_1$, and so on. Packaging t steps as A^t comes later. To build A from a flow, ask: if everything starts in compartment j, where does it go next? Those destinations are column j.

Worked in class

W1. Classify from RREF. Unique, none, or infinitely many?

(a)
$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$$

(b)
$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right]$$

(c)
$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

W2. Parametric form. From

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right],$$

name the free variable and write $x = (x_1, x_2, x_3)$ in terms of a parameter.

W3. Build a transition matrix. Two towns, Arcata and Eureka, exchange residents each year:

- Of those in Arcata, 80% stay and 20% move to Eureka.
- Of those in Eureka, 90% stay and 10% move to Arcata.

Order the state as $x_t = (\text{Arcata}, \text{Eureka})$. Write A. With $x_0 = (5000, 7000)$, compute x_1. Does Arcata rise or fall in that year?

W4 (if time). State and matrix. A campus tracks (on-campus, off-campus) as x_t with

$$A = \begin{bmatrix} 0.7 & 0.15 \\ 0.3 & 0.85 \end{bmatrix}.$$

(a) What does A_{12} mean in words? (b) What does the first column represent? (c) If $x_0 = (800, 200)$, compute x_1 and interpret both entries as people.

W5 (if time). One residual. For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 11 \end{bmatrix},$$

solve $Ax = b$ with Octave or Python and compute $Ax - b$.

Your turn

Start with Y1–Y3 and Y5–Y6. The others are extra practice of the same skills.

Y1. From the RREF

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

list pivot variables, free variables, and the general solution.

Y2. True or false: if RREF has a row of all zeros including the augmented entry, the system is inconsistent.

Y3. A 3×5 matrix A, after a consistent RREF of $[A \mid b]$, has 3 pivots. How many free variables? Unique solution or

infinitely many?

Y4. You solve $Ax = b$ in software and get $x = (1, -1)$. You compute $Ax - b = (0.0,\ 10^{-15})$. Was the solve successful? Why check the residual?

Y5. True or false, with a one-line reason: if $x_{t+1} = Ax_t$, then knowing x_t and A determines x_{t+1} uniquely.

Y6. Two warehouses, $x_t = (\mathrm{A}, \mathrm{B})$. Each day, A keeps 70% and ships 30% to B; B keeps 60% and ships 40% to A. Write A. With $x_0 = (100, 50)$, find x_1.

Y7. With $A = \begin{bmatrix} 1.1 & 0 \\ 0 & 0.5 \end{bmatrix}$ and $x_0 = (2, 8)$, compute x_1 and x_2. Which component is growing? Which is shrinking?

Y8 (if time). Three age groups (youth, adult, senior):

$$A = \begin{bmatrix} 0.6 & 0 & 0 \\ 0.4 & 0.7 & 0 \\ 0 & 0.3 & 0.8 \end{bmatrix}, \qquad x_0 = (100, 200, 50).$$

Interpret A_{21} and A_{33} in one phrase each, and compute x_1.

Answers (Your turn)

Y1. Pivot variables x_1, x_2; free $x_3 = t$. Then $x_1 - 2t = 5$ and $x_2 + 3t = -1$, so

$$x = (5 + 2t,\, -1 - 3t,\, t), \quad t \in \mathbb{R}.$$

Y2. **False.** A zero row including the right-hand side is consistent (a redundant equation). Inconsistency needs $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$.

Y3. $n = 5$ unknowns and 3 pivots $\Rightarrow$ 2 free variables $\Rightarrow$ infinitely many solutions (when consistent).

Y4. Yes — the residual is numerically zero (roundoff). Checking $Ax - b$ catches a wrong x, a singular system, or a copy-paste mistake. Fuller software practice is Lab 3.

Y5. **True.** The rule $x_{t+1} = Ax_t$ is a function of the current state: one matrix–vector product gives the unique next state.

Y6.

$$A = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}, \qquad x_1 = \begin{bmatrix} 90 \\ 60 \end{bmatrix}.$$

Column 1: keep 70% in A and ship 30% to B. Column 2: ship 40% of B to A and keep 60% in B.

Y7. $x_1 = (2.2, 4)$, $x_2 = (2.42, 2)$. The first component grows (factor 1.1 each step); the second shrinks (factor 0.5).

Y8. $A_{21} = 0.4$: the factor sending youth to adults in one step. $A_{33} = 0.8$: the factor of seniors who remain seniors.

$$x_1 = Ax_0 = \begin{bmatrix} 60 \\ 180 \\ 100 \end{bmatrix}.$$